

Bolt Details:

Size; M6

Grade; 12.9

Type; Socket Head Cap Screw

No. of Bolts; $N_b = 8$

Tensile Strength; $S_t = 1220 \text{ MPa}$

Tensile Stress Area; $A_t = 20.1 \text{ mm}^2$

Load Details:

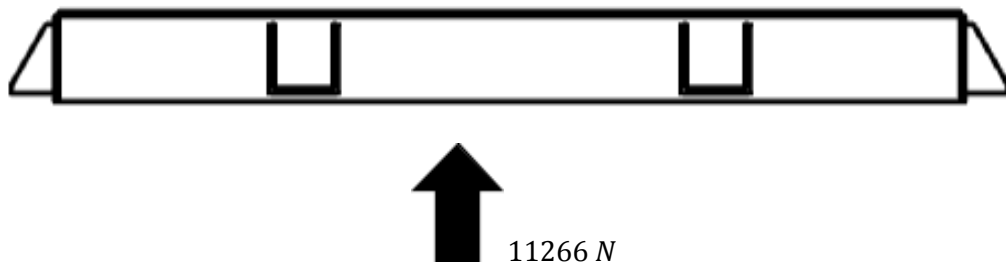
The Max Load Experienced is 20 G in the vertical direction.

Segment Mass; $m_s = 6.38 \text{ kg}$

No. of Segments; $N_s = 9$

Total Segment Mass; $m_t = 57.42 \text{ kg}$

Load; $F = 20 * 9.81 * m_t = 20 * 9.81 * 57.42 = 11266 \text{ N}$



$$\sigma = \frac{F}{A_t * N_b}$$

$$\sigma = \frac{11266}{0.000201 * 8}$$

$$\sigma = 7006219 \text{ Pa}$$

$$\sigma = 7.01 \text{ MPa}$$

$$F_s = \frac{S_t}{\sigma}$$

$$F_s = 174.04$$